

Lekkala Ganesh

AI Application Engineer | Full-Stack Engineer

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SUMMARY

AI application engineer building **Python** and **TypeScript** systems that put machine learning into production: **embedding-based retrieval (RAG)**, **scikit-learn** text matching, **ONNX** model serving, and **multi-provider LLM orchestration** with fallback — wrapped in **FastAPI** services, **PostgreSQL/SQL** schema design, and React clients. **320+ automated tests** and **CI** across public repos. B.Tech CSE, seeking AI/ML engineering roles.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

Machine Learning: scikit-learn, Pandas, NumPy, Regression, Classification, Feature Engineering, Cross-Validation, Hyperparameter Tuning, Model Evaluation, EDA, TF-IDF, Cosine Similarity

AI & Deep Learning: PyTorch, Hugging Face (Diffusers, Transformers), sentence-transformers, ONNX Runtime, OpenCV DNN, 4-bit NF4 Quantization (bitsandbytes), CUDA

Generative AI & RAG: GPT-4o, Gemini, Ollama (local inference), Embeddings, Vector Search (Milvus, Pinecone), Semantic Search, Prompt Engineering, Multi-Provider Fallback, SSE Streaming

Backend & Data: FastAPI, Flask, Node.js, REST APIs, WebSockets, Async Job Queues, Pydantic, PostgreSQL (Row-Level Security, Triggers, Indexes), SQLite, Supabase, MongoDB, Schema Design, Joins & Aggregations, Parameterized Queries

Frontend, Testing & DevOps: React, Next.js (SSR/SSG), Tailwind CSS, pytest, Vitest, Playwright, GitHub Actions (CI/CD), Docker, Git, Linux, Vercel

EXPERIENCE

Freelance Full-Stack & AI Engineer

Remote

Self-Employed

Jan 2025 – Present

- **LLM Integration:** Engineered **multi-provider AI fallback** (GPT-4o → Gemini → OpenRouter) with graceful degradation, powering real-time **SSE** voice streaming (**VoxAI**) and 3D sunlight simulation.
- **Data Modeling:** Designed **PostgreSQL** schemas — foreign keys, CHECK constraints, composite indexes, triggers — securing per-user data with **Row-Level Security** policies and **JWT** auth on Supabase.
- **Full-Stack Delivery:** Shipped client web apps end to end — **REST API** design and component-driven React/Next.js front-ends meeting **WCAG 2.1 AA**, with automated **Playwright** cross-browser testing.

Full Stack Web Development Intern

Remote, Part-time

Slash Mark IT Solutions

Dec 2024 – Apr 2025

- Delivered **6 full-stack projects** across 3 tiers: real-time Weather Forecast, URL Shortener API (Node.js/Express/MongoDB), and a Food Delivery platform with auth, geolocation, and payment flow.

Machine Learning Intern

Virtual

Bharat Intern

Oct 2023 – Nov 2023

- Built **regression** and **classification** models (House Price, Wine Quality, Iris) in **Python** with **scikit-learn** and **Pandas** — **data cleaning**, **EDA**, and **feature engineering**; compared estimators via **train/test splits**, **cross-validation**, and **hyperparameter tuning** on held-out **evaluation** metrics.

PROJECTS

ATS Resume Optimizer | *Python, scikit-learn, Streamlit, SQLite, GitHub Actions*

[GitHub](#)

- Resume-to-job **semantic matching** via **scikit-learn TF-IDF** and **cosine similarity** with keyword-gap extraction; **269** **pytest tests** and **GitHub Actions CI** (3.11/3.12, ruff) over **~20K lines of Python**.

AI Interview Trainer | *FastAPI, Ollama, NumPy, SQLite, Next.js, Docker*

[GitHub](#)

- Local-first **RAG memory** storing **embeddings** as SQLite BLOBs with hand-implemented **cosine-similarity retrieval**; tiered **Ollama** routing with fallback, **SSE** streaming, **53** **pytest tests**, Dockerized.

Lite-Vision — Real-Time Face Analysis | *FastAPI, OpenCV DNN, ONNX, Next.js*

[GitHub](#)

- Serves **four pretrained ONNX models** (SCRFD, InsightFace gender-age, FairFace, FER+) via an **OpenCV DNN** pipeline with **multi-model fusion**, CLAHE preprocessing, and frame caching; rate-limited **FastAPI**, live webcam streaming, **73 tests**.

AuraGen — AI Image Generation Platform | *PyTorch, HF Diffusers, FastAPI, CUDA*

[GitHub](#)

- **Async job queue** with **WebSocket** progress streaming and OOM handling; **hybrid cloud-burst routing** offloads jobs over the local **VRAM budget** to HuggingFace with local fallback; **NF4 quantization** path (bitsandbytes, attention slicing, VAE tiling).

EDUCATION

Parul University

Bachelor of Technology in Computer Science & Engineering; GPA: 7.53/10

Vadodara, Gujarat

Jul 2022 – Jun 2025